

A Direct Approach to Optimal Feeder Routing for Radial Distribution System

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Abstract—Much of the research has been focused on developing the optimization techniques, varying from classical to nontraditional soft computing techniques, to solve the distribution system planning problem. Most of the methods preserve the distinctions and niceties, but dependent on complex search algorithm with a lot of convergence related issues that require more time to reach a firm conclusion at planning stage. This paper proposes a simple direct solution that significantly reduces the inherent difficulties of finding the solution and ensures optimum solution at the same time. Moreover, the concept of principle of optimality is effectively used to make the proposed technique more computationally efficient and useful. The effectiveness of the developed planning technique has been verified with different test cases.

Index Terms—Distribution system planning, direct solution approach, principle of optimality, load flow analysis.

I. INTRODUCTION

THE main objective of the distribution system planning is to determine the substation location, size and its service area, number of feeders and their routes. A planning cost and its constraints are accordingly formulated to obtain the solution. Mathematically, the problem is nonlinear, non-differentiable, and combinatorial in nature with a large number of discrete and continuous variables. The problem is solved in different time frames as either single or multiple horizons employing the right strategy. Various mathematical programming techniques, such as integer programming, mixed integer programming, transportation, transshipment, and branch and bound methods have been applied to the problem of feeder routing optimization [1]–[12]. Among the classical solution methods, the most preferred method is mixed integer linear programming (MILP) [2], [6]–[9]. The MILP is a multistage solution technique where the nonlinear problem is transformed to linear problem without any integer restriction and is solved by employing any linear programming technique. The variables are then modified by the commonly used branch and bound method. These models are difficult to solve and the solution time increases for such problem like distribution system planning, where large number of discrete variables are present. Evidently good initial guess of the variables can reduce the solution time [8], [9]. The planning of secondary distribution circuits is also approached as a mixed integer nonlinear programming problem (MINLP) [10]. A

dedicated Evolutionary Algorithm (EA) is proposed to solve this problem. Fletcher and K. Strunz proposed a mathematical model for optimal expansion of both primary and secondary distribution systems for each period of the planning horizon [11], [12]. That model is solved using a classical optimization technique based on a direct derivative gradient search method. Antonio Marcos Cossi *et al.* formulated the secondary distribution circuit expansion as a MINLP and Tabu search method is used to solve the problem [13].

Recently, some works using genetic algorithms (GAs), Simulated Annealing, and Ant Colony based techniques for planning of distribution system have been reported [14]–[19]. Besides obtaining higher chance to achieve better result, nonlinearity of the cost function, real and integer variables, nonlinear constraints can easily be incorporated in formulation while using these optimization techniques [16]–[18]. The potential of the GA is shown in comparison with the classical technique [16]. This is true that in GA, the length of binary strings representing the integer variables becomes high even for medium size network and therefore, this leads to extremely large number of unfeasible solutions. Simulated Annealing technique is applied to solve the problem [17]. Author in this Simulated Annealing based approach generates initial solution by steepest descent method. Then the initial network is modified by including a new branch that is selected randomly. Subsequently the newly formed mesh is opened by removing a branch selected by a random selection process again. Each of the above moves follows a full load flow to check the effect of inclusion of new branch in the network. In the Ant Colony based method, the optimal network is obtained by movement of ants from one node to another node where each move is decided by magnitude of pheromone accumulated in different paths and a heuristic guide function. A number of parameters are required to tune and 1000 runs of the algorithm are required for proper tuning of parameters, as reported by Gomez *et al.* [18]. A different approach based on branch-exchange techniques have been applied in distribution system planning [20], [21]. A more detailed review of different models is available in [22].

The motivation behind the work is to reduce the complexities associated with the methodology of distribution system planning without compromising the optimality of the solution. Therefore, a comprehensive approach of radial distribution planning has been developed involving all the associated parameters and possible constraints.

II. OBJECTIVE FUNCTION

Distribution system planning analysis is designed to minimize the total annual cost which is the summation of the fixed cost, energy cost and interruption cost [17].

The yearly fixed cost recovery is given as

$$C_f = g \sum_{k \in M} C_k \quad (1)$$

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where C_k is the cost of branch k of the main feeder and g is the yearly recovery rate of fixed cost. Costs of branches emanating from the source substation include both the line and the corresponding substation costs. M denotes the set of all possible branches in a particular radial path in the network configuration under consideration.

The second component is the cost of energy losses that increases many folds during the entire life of the network if the network is selected other than the optimum one. This cost of energy losses " C_l " may be represented as

$$C_l = 8760p\beta \sum_{k \in M} (I_k)^2 r_k \quad (2.a)$$

$$\beta = 0.15\alpha + 0.85\alpha^2 \quad (2.b)$$

where

- p cost per unit of energy lost;
- β loss factor;
- r_k branch resistance;
- I_k branch current at peak load;
- α load factor.

The third component, interruption cost, becomes high for a radial path when there is no power available in entire route for any interruption. In the radial networks, there is no alternative supply route and the outage of a branch interrupts the delivery to all the consumers supplied through this branch. Hence, the cost of outage can be calculated using the following expression (3):

$$C_i = c_i \alpha d \sum_{k \in M} \lambda_k \text{Re}\{I_k\} \sqrt{3} U_r \quad (3)$$

where

- c_i cost per unit of energy not delivered;
- α load factor;
- d repair duration;
- λ_k branch failure rate;
- U_r network rated voltage;
- I_k branch current at peak load.

The total annual cost that should be minimized is

$$C = C_f + C_l + C_i. \quad (4)$$

The constraints to be satisfied are:

- i) Conservation of power flow: $\overline{TP} = \overline{D}$ where $\overline{P} = (P_1, P_2, P_3, \dots, P_N)$ is the vector of power flow, and $\overline{D} = (D_1, D_2, D_3, \dots, D_N)$ is the vector of power demand at each of N nodes, and I is node element incidence matrix.
- ii) Capacity constraint $P \leq U$, U is the vector of capacity limits. Maximum current limit in case of feeder, maximum practical kVA rating in case of substation.
- iii) Radiality constraint.
- iv) Voltage drop constraint.

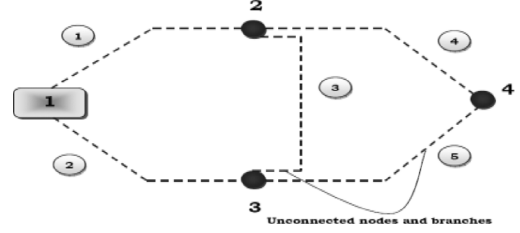


Fig. 1. Available supply routes for four nodes.

III. FORMATION OF OPTIMUM RADIAL NETWORK

The proposed direct solution technique is basically based on searching the optimum path for energizing a node among all the possible paths. There may be several possible radial paths to reach a node, from a substation. The minimum cost path among all the radial paths for feeding a particular node will be the optimum path for the node. This essentially is an iterative search technique where the minimum path for a node is searched in iteration. In the beginning of the search, all possible radial paths for each node have been traced out. As in Fig. 1, 1-2-3-4 is one of the four different available radial paths to feed a node 4. Forward/backward load flow technique is then applied to find the energy loss and interruption cost. Subsequently planning cost is calculated for each of the four radial paths according to (4).

There are basically two problems to be solved. The first one involves tracing out the radial paths from large number of possible connections and the second one is computing the cost of calculation for large number of radial paths. An easy step by step algorithm is proposed to solve the first problem. The concept of the principle of optimality is used for the second problem. The principle of optimality theorem states that from any point on an optimal trajectory, the remaining trajectory is optimal for the corresponding problem initiated at that point [23]. There may be more than one node on any optimum path. Once an optimal path is found for any node, the theorem states that the parts of this optimal path will be the optimal path for all the nodes which are lying along the trajectory. Therefore, it is not required to carry out searching the optimum radial path for all the nodes separately. As a result, cost calculation is required for making decision on limited nodes only.

Global solution is ensured by verifying all the possible radial paths to reach a node before deciding the optimum route for individual node.

A. Algorithm for Tracing Radial Paths for All the Nodes

The radial distribution system is always a directed path, where power is directed from a substation to the load nodes. A directed graph is an ordered pair $D = (V, A)$ with $V =$ a set whose elements are called vertices or nodes, and $A =$ a set of ordered pair of vertices, called arcs, directed edges or arrows. An arc $A = (x, y)$ is considered to be directed from x to y , where y is called head and x is called the tail of the arc. Every arc will be a radial path starting from any substation.

Let us consider a network as shown in the Fig. 1 with all possible paths to feed the three load nodes (2, 3, 4) from substation node 1. The dotted lines represent the probable connections to energize the nodes from the substation node 1. A step by step radial path building algorithm developed to determine all the radial paths is summarized below.

STEP 1: Initial arcs with substation

This work initiate with development of arc with the load nodes

connected directly with substation. Therefore, the number of initial arcs depends on the number of feeders coming out from the substation. In an arc (x, y) , x (tail) will be the substation and y (head) will be the set of load nodes that are connected with x . $Y = \{y_1, y_2, \dots, y_n\}$, where, y = set of load nodes directly connected with substation x . Thus there are n numbers of arcs after first step that can be written as

$$A = \{(x y_1), (x y_2), \dots, (x y_n)\}.$$

Referring to the Fig. 1, we can write

$$A = \{(1 2), (1 3)\} \quad (5)$$

where x = Substation node 1, y = A set consisting of load nodes 2 and 3 which are directly connected with the substation node 1.

STEP 2: New arcs by extending load nodes

At the next step, load nodes connected with substation node acts as tail(x) of the new arc where in the new head(y) depends on the connectivity with the other nodes. As explained earlier, an arc represents a radial path and therefore a new arc adds only a head at a time with an old arc. While updating an arc, it is to be checked that head does not appear in an existing or old arc. Updating of arcs will be continued till all the possible load nodes are covered. As in (5), nodes 2 and 3 will now be acting as tails in step 2, to be connected with new heads. From Fig. 1, it is clear that 2 may be connected to 1, 3 or 4. But 1 is already present in initial arcs; therefore 1 will not be appearing in a new arc. With node 2, updated arcs are $\{(1 2 4), (1 2 3)\}$. Similarly, with node 3, updated arcs are $\{(1 3 2), (1 3 4)\}$. Hence, New updated $A = \{(1 2), (1 3), (1 2 4), (1 2 3), (1 3 2), (1 3 4)\}$. The continuation of this update will happen till all the four nodes are expanded. Combining all arcs, the final A will be, $\{(1 2), (1 3), (1 2 4), (1 2 3), (1 3 2), (1 3 4), (1 2 4 3), (1 2 3 4), (1 3 2 4), (1 3 4 2)\}$.

STEP 3: Zero padding

In this step, set A is converted to equivalent matrix E of specific dimension. In order to maintain same number of elements in each row, zero padding is carried out. Hence, the number of zeros to be entered in each row is calculated using (6)

$$\text{Number of zeroes in each row, } Z_p = [(n + 1) - N_z] \quad (6)$$

where n = total number of load nodes and N_z = non zero elements of a radial path.

In matrix form, we can write

$$E = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 2 & 4 & 0 \\ 1 & 2 & 4 & 3 \\ 1 & 2 & 3 & 0 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 2 & 0 \\ 1 & 3 & 2 & 4 \\ 1 & 3 & 4 & 0 \\ 1 & 3 & 4 & 2 \end{bmatrix}. \quad (7)$$

If there is more than one substation, the same steps 1 to 3 will be followed to construct final E matrix for each substation. It is apparent from (7), that the last nonzero element in each row of matrix E represents the last load node of its radial path. As in

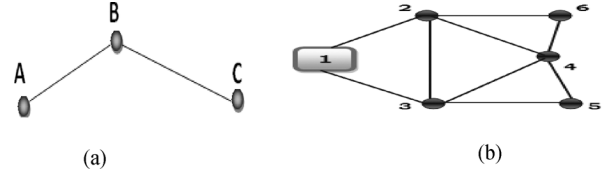


Fig. 2. (a) A radial path of three nodes. (b) Available supply routes for 5 load nodes distribution system.

(7), node 2 appears three times as last nonzero nodes, therefore, there are three different possible radial paths to energize node 2, as represented by a matrix E_2 in (8)

$$E_2 = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 3 & 2 & 0 \\ 1 & 3 & 4 & 2 \end{bmatrix}. \quad (8)$$

After that, we have to calculate the cost of the each radial paths of matrix E_2 , applying (4). The minimum cost path is selected as optimum for node 2.

B. Principle of Optimality

Cost function is to be evaluated for a number of radial paths before finalizing the route to energize a node. The strategy becomes computationally overburdened considering the total number of paths arising for a given node. Computational complexity can be drastically reduced and consequently the strategy becomes more effective if the theorem of principle of optimality is followed. The theorem is largely concerned with dynamic programming optimization technique which states that: If the best possible path from A to C passes through intermediate point B, then the best possible path from B to C must be the corresponding part of the best path from A to C as in Fig. 2(a).

Referring to Fig. 2(b), if 1-2-4-6 is found to be optimal for node 6, then according to optimality theorem there is no need to check the optimal path for nodes 4 and 2 as these nodes are on the optimal trajectory because 1-2-4 and 1-2 are the parts of optimal path 1-2-4-6. Thus this strategy of using principle of optimality reduces searching of optimal paths.

C. Proof of Principle of Optimality in the Context of Radial Network

Let us take an example of four bus to prove the principle of optimality for planning a distribution system having two paths available for energizing node 6 as shown in Fig. 3(a) and (b). After calculating the annual costs of these two routes using (4), it is found that

$$TC_1^6 < TC_2^6 \quad (9)$$

where

TC_1^6 cost of the whole path of the 1-2-4-6;

TC_2^6 cost of the whole path of 1-3-4-6.

Therefore, path 1-2-4-6 is the optimal trajectory for node 6. According to the optimality theorem, for energizing node 4 path 1-2-4 as shown in Fig. 3(c), will be the optimum rather than path 1-3-4 as shown in Fig. 3(d). This is because 1-2-4 is the part of the already established optimal trajectory of 1-2-4-6. Therefore, in accordance with the theorem, 1-2-4 should be optimum and the following relation should not be true:

$$TC_1^4 > TC_2^4 \quad (10)$$

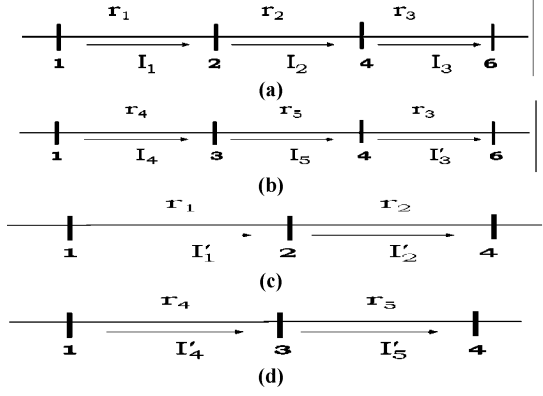


Fig. 3. Four bus example.

Where, TC_1^4 is the total cost to energize node 4 by the route of Fig. 3(c) and TC_2^4 is the total cost to energize the same node 4 by the route of Fig. 3(d). For energizing node 4, the total cost of the path of Fig. 3(c) is

$$TC_1^4 = \sum_{i=1}^{nb} C_{f_i} + \sum_{i=1}^{nb} C_{l_i} + \sum_{i=1}^{nb} C_{i_i} \quad (11)$$

and the same for Fig. 3(d),

$$TC_2^4 = \sum_{i=1}^{nb} C_{f_i} + \sum_{i=1}^{nb} C_{l_i} + \sum_{i=1}^{nb} C_{i_i} \quad (12)$$

where nb is the total branches of the respective figures. TC_1^6 can be written as the summation of TC_1^4 calculated from (11) and the extra fixed cost (ΔC_f), variable cost (ΔC_l) and interruption cost (ΔC_i) for adding new branch 4–6 as

$$TC_1^6 = TC_1^4 + \Delta C_{f_1} + \Delta C_{l_1} + \Delta C_{i_1}. \quad (13)$$

A similar equation is written for the same to feed 6 by another route of Fig. 3(b) as

$$TC_2^6 = TC_2^4 + \Delta C_{f_2} + \Delta C_{l_2} + \Delta C_{i_2}. \quad (14)$$

ΔC_{f_1} and ΔC_{f_2} appeared in (13) and (14) and are equal as the extra fixed cost is added for the same branch 4–6. But the changes in variable and interruption costs for adding 4–6 may not be the same as the currents I_3 and I'_3 flowing in branch 4–6 as indicated in Fig. 3(a) and as Fig. 3(b) are dependent on the respective voltages at bus 4. Considering the worst condition for optimum path 1-2-4-6 of Fig. 3(a), when both $\Delta C_{l_1} > \Delta C_{l_2}$ and $\Delta C_{i_1} > \Delta C_{i_2}$, then from (13) and (14),

$$TC_1^6 > TC_2^6, \quad \text{if } TC_1^4 > TC_2^4$$

which stands for cost of 1-2-4-6 > cost of 1-3-4-6 if cost of 1-2-4 > cost of 1-3-4. But path 1-2-4-6 has already been found optimum for node 6 as in (9). Hence, $TC_1^4 > TC_2^4$ will never be true. Consequently, it can be concluded that 1-2-4 is the optimal path for node 4 as it is a part of optimal trajectory of 1-2-4-6.

D. Effective Way of Applying the Technique

As per the principle of optimality, separate search is not required to identify the optimal path for a given node, once this

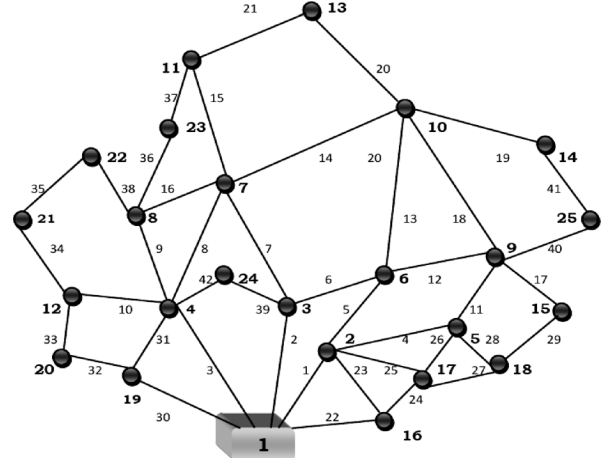


Fig. 4. Graph of available supply routes for 24 load nodes distribution network.

TABLE I
ZERO PADDING FOR THE RADIAL PATHS OF FIG. 1

Each radial path of A	Corresponding row after zero padding
{1 2}	[1 2 0 0]
{1 3}	[1 3 0 0]
{1 2 4}	[1 2 4 0]
{1 2 3}	[1 2 3 0]
{1 3 2}	[1 3 2 0]
{1 3 4}	[1 3 4 0]
{1 2 4 3}	[1 2 4 3]
{1 2 3 4}	[1 2 3 4]
{1 3 2 4}	[1 3 2 4]
{1 3 4 2}	[1 3 4 2]

node is on the optimal path of any other node. Therefore, to obtain maximum advantage from the principle of optimality theorem, the optimal routing should be started with the terminal nodes that are farthest away from the substation. As per Fig. 4, terminal nodes to be considered are 13, 10, 14, 25, 15, 11, 22, and 21. The identified optimal paths of these terminal nodes are likely to cover most of the other nodes of the system on their paths. Separate search is only required for the nodes which are not covered on the optimal paths identified for the terminal nodes. Therefore, there is no chance of missing the optimum number of main feeders, their routings and routing of their respective laterals. The computational time is also drastically reduced as cost evaluations need to be done for fewer paths. Detailed discussion is presented in test case-I under Section V.

IV. DETERMINATION OF OPTIMAL NUMBER OF SUBSTATION

Substation number and locations are also needed to be optimized in case, when multiple optional substation locations are available in a distribution planning problem. Firstly we select optimal feeder routes to feed each load node to an available optional substation node. The proposed direct method generates the solution of connectivity of load nodes connected through any one of the proposed substations. Hence, different substation zones are formed with a substation node and associated load nodes connected to it. Thus, each substation zone comprises the substation and the loads connected with the substation. The cost per kVA of each substation zone is then calculated. Total cost for a zone includes the cost of substation, cost of the feeders and the cost of energy loss. Cost per kVA is then calculated by dividing

this cost by the actual connected load of the zone. Thus we get (15), as shown at the bottom of the page, where

A_c cost of the substation;

A_f cost of the feeder;

A_e cost of the energy loss.

Based on the calculated cost per kVA for an individual substation, the optimal number of substations is determined according to following steps.

- Planning with all probable substations.
- Eliminate the substation whose cost per kVA is the highest among all probable substations
- Planning is performed with the remaining substations.
- Reject the substation with the highest cost per kVA and repeat until the total capacity of substations is sufficient to meet the full load demand with safe capacity margin.

V. SIMULATION RESULTS

The feasibility of the developed planning technique has been verified by two test cases.

A. Test Case-I

The distribution planning problem reported in [17] has been considered. The system has 25 nodes and 42 available branches for their supply from the substation 35 kV/10.5 kV at node 1, as shown in Fig. 4. Total load in the network is 2.55 MVA. The substation cost has been included with the feeder related cost for every outgoing line emanating from substation is 75 k\$.

As the first step, the matrix “E” as in (7) containing all the possible radial paths is framed following the steps of building algorithm. Finally building algorithm generates 5589 total radial possible paths for energizing all nodes as shown in column-3 of Table II.

After tracing all the possible radial paths for energizing all nodes, decisions for finding optimal routes are taken for the nodes 13, 10, 14, 25, 15, 11, 22 and 21. At the beginning of the method, any node can be selected but it is recommended to select those nodes first which are farthest from substation to reduce the cost evaluation burden as well as time required. Further search is not required to find the optimal path for any node, once this node gets a place on any optimal trajectory of feeding any other node. Decisions are required for only 7 nodes that are sufficient to cover all the other nodes of the system which is also apparent from Table II. For example, the optimal path for 13 decides the routes of energizing nodes of 19, 4, 8, 3, 11, and 13. As in Table II, all the other loads are lying on at least one optimal trajectory found for terminal nodes except 3 and 7. Therefore, cost calculation for the radial paths reduces to 1580 from the total paths of 5589 for effectively using the of optimality theorem.

The proposed technique is developed on MATLAB R2006b code and the computational time required is only 5 minutes

TABLE II
OPTIMAL PATHS FOR FEEDING TO ALL 24 LOAD NODES

Node to be energized.	Optimal path.	Total available radial paths for the energized node.
15	1-16-17-5-18-15	163
14	1-16-17-5-9-25-14	188
10	1-16-2-6-10	319
13	1-19-4-8-23-11-13	172
22	1-19-20-12-21-22	166
17	1-16-17	224
18	1-16-17-5-18	180
9	1-16-17-5-9	354
25	1-16-17-5-9-25	182
11	1-19-4-8-23-11	231
23	1-19-4-8-23	178
8	1-19-4-8	311
21	1-19-20-12-21	176
12	1-19-20-12	206
20	1-19-20	192
7	1-19-4-7	326
5	1-16-17-5	261
6	1-16-2-6	292
24	1-19-4-24	196
16	1-16	178
19	1-19	210
2	1-16-2	300
4	1-19-4	338
3	1-19-4-24-3	246
Total radial paths for all nodes		5589
Total shaded paths		1580

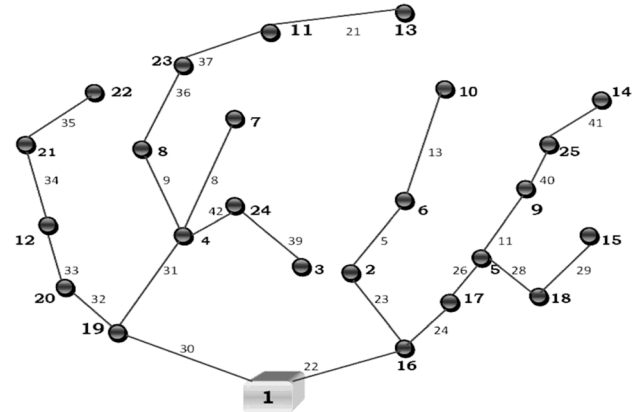


Fig. 5. Optimal network structure of the 24 load nodes distribution network.

TABLE III
ANNUAL NETWORK COST IN US\$ FOR TEST CASE I

Optimal solution	C_f	C_l	C_i	TOTAL COST
	25500	17287	1408	44195

using computer: PC Pentium, 1.98 GHz, and 1 GB of RAM. Total losses occurring in this network amounts to 0.0314 MW. Table III shows the total cost and the individual cost component of the cost function for test case-I. Optimal network structure obtained by this method is shown in Fig. 5, which is similar to the network obtained from the reported Simulated Annealing method [17].

$$\text{Cost per kVA for each substation} = \frac{A_c + A_f + A_e}{(\text{Total kVA of the connecting load with the substation})} \quad (15)$$

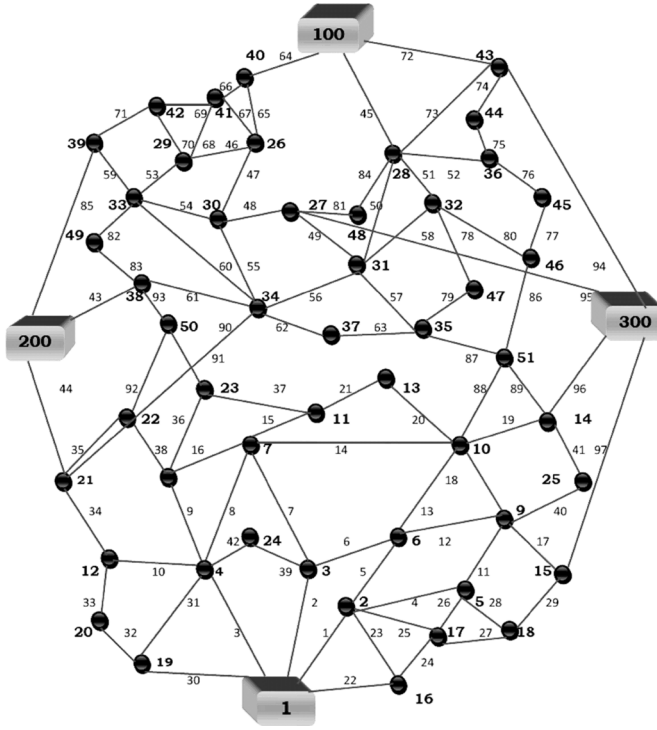


Fig. 6. Graph of available supply routes for Test Case-2.

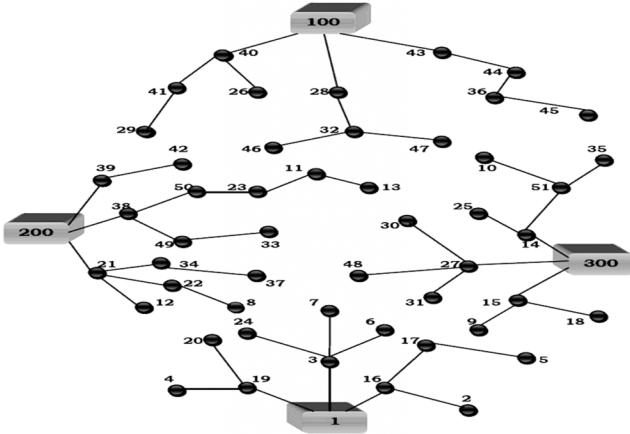


Fig. 7. Optimum network with four substations.

Despite dependency on cooling schedule, the proposed technique of Simulated Annealing generates solution around current point by bringing a randomly selected branch. In Ant colony and GA based method, tuning of several critical parameters affects convergence. The minimum parameters required to tune are five and seven in GA and Ant Colony respectively [16], [18]. The proposed direct approach method does not require any tuning of any parameter for obtaining solution. Moreover, while making the decision regarding the optimal path, all the possible radial paths have been checked, which the solution ensures the best or global optimum path for any node.

B. Test Case-II

A comparatively large network with 4 substations, 50 load nodes, and possible 97 branches are considered which is shown in Fig. 6. The details on consumption at load points, length of graph branches, line data and load data are depicted in Appendix A. The total load connected in this network is 5.3

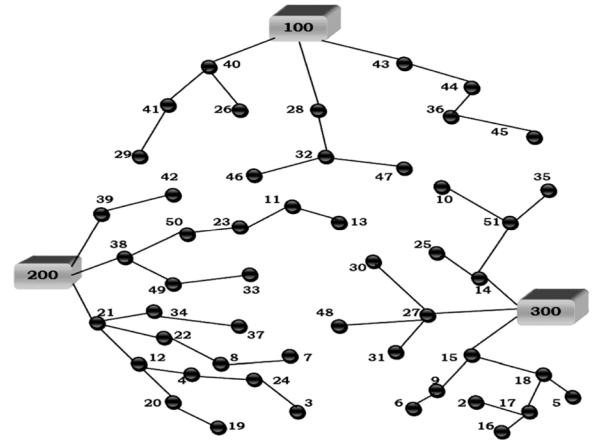


Fig. 8. Optimum network with three substations.

TABLE IV
COST PER KVA FOR EACH SUBSTATION OFF ALL COMBINATIONS

COMBINATIONS	INDIVIDUAL SUBSTATION'S COST(\$/kVA)				TOTAL COST (\$)
	1	100	200	300	
When all four substations are present in Fig. 7	61.33	60.00	56.86	60.42	31.3256×10^4
When 1 is rejected for high cost/kVA in Fig. 8	-	60.00	55.74	56.24	30.1787×10^4
When 100 is rejected for high cost/kVA in Fig. 9	-	-	54.09	48.49	27.075×10^4
When 200 is rejected for high cost/kVA	-	-	-	58.48	28.998×10^4

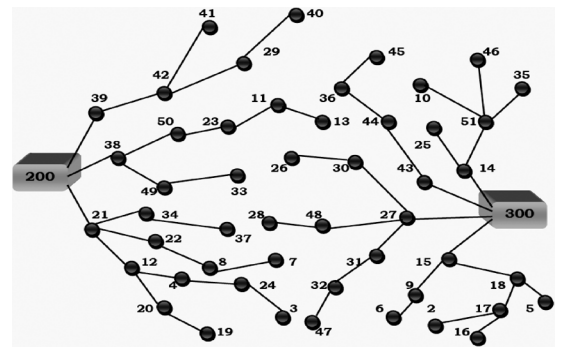


Fig. 9. Optimum network with two substations.

MVA and power factor is assumed to be 0.9. Substation 1 which produces highest cost per kVA i.e., 61.33 is rejected and the network shown in Fig. 8 is optimal with remaining three substations. Removal of substation 1 is justified as the network cost is reduced to 30.1787×10^4 \$ from 31.3256×10^4 \$ as shown in Table IV.

Table IV also shows that a network with substations 200 and 300, shown in Fig. 9, comes out as minimum cost configuration after following the rejection process of higher cost/kVA substation. Table V shows that variable cost and interruption cost are

TABLE V
ANNUAL NETWORK COST IN US\$ FOR TEST CASE II

COMBINATIONS	C_f	C_l	C_i	TOTAL COST
When all four substations are present.	288800	27839	2662	31.3256×10^4
When 1 is rejected for high cost/kVA	266887	32025	2869	30.1787×10^4
When 100 is rejected for high cost/kVA	231180	36525	3053	27.075×10^4
When 200 is rejected for high cost/kVA	243200	42970	3809	28.998×10^4

becoming high with the removal of substations, though the fixed costs are becoming less, except for the last one.

VI. CONCLUSION

This approach builds on simple concept that depends solely on tracking of radial paths and calculating cost associated with the paths. An easy step by step building algorithm is proposed to find the radial paths existing in a system. The work has also explored the advantage of theorem of principle of optimality and successfully established a link with radial system planning that makes the process of searching faster and more effective. The proposed technique ensures global optimal solution without being influenced by initial paths as considered in classical techniques or different parameters as required by soft computing techniques for tuning. Along with the routing, the solution of optimal number of substations is also proposed. Radial path building algorithm may be considered a useful tool for typical route planning problem.

APPENDIX A

DATA FOR TESTCASE-I (REFERRED TO REF. [17]) DATA FOR TESTCASE-II

TABLE A.I
CONSUMPTION AT LOAD POINTS

Load point no	2	3	4	5	6	7
Load, kVA	250	160	100	100	50	100
Load point no	8	9	10	11	12	13
Load, kVA	100	250	160	100	160	100
Load point no	14	15	16	17	18	19
Load, kVA	100	100	150	80	40	100
Load point no	20	21	22	23	24	25
Load, kVA	40	60	40	80	100	30
Load point no	26	27	28	29	30	31
Load, kVA	250	160	100	100	50	100
Load point no	32	33	34	35	36	37
Load, kVA	100	250	160	100	160	100
Load point no	38	39	40	41	42	43
Load, kVA	100	100	150	80	40	100
Load point no	44	45	46	47	48	49
Load, kVA	40	60	40	80	100	30
Load point no	50	51				
Load, kVA	100	100				

TABLE A.II
LENGTH OF GRAPH BRANCHES

Branch no.	1	2	3	4	5	6
Length, km	2.10	1.65	2.20	2.00	1.50	1.75
Branch no.	7	8	9	10	11	12
Length, km	1.75	1.75	1.00	1.00	1.25	1.50
Branch no.	13	14	15	16	17	18
Length, km	1.75	2.00	2.00	1.75	1.25	1.75
Branch no.	19	20	21	22	23	24
Length, km	1.75	2.75	1.75	1.50	1.05	0.75
Branch no.	25	26	27	28	29	30
Length, km	1.05	1.00	1.50	0.75	1.25	1.55
Branch no.	31	32	33	34	35	36
Length, km	1.00	0.75	0.75	0.50	0.50	1.05
Branch no.	37	38	39	40	41	42
Length, km	0.50	0.65	0.75	0.45	0.50	0.40
Branch no.	43	44	45	46	47	48
Length, km	2.10	1.65	2.20	2.00	1.50	1.75
Branch no.	49	50	51	52	53	54
Length, km	1.75	1.75	1.00	1.00	1.25	1.50
Branch no.	55	56	57	58	59	60
Length, km	1.75	2.00	2.00	1.75	1.25	1.75
Branch no.	61	62	63	64	65	66
Length, km	1.75	2.75	1.75	1.50	1.05	0.75
Branch no.	67	68	69	70	71	72
Length, km	1.05	1.00	1.50	0.75	1.25	1.55
Branch no.	73	74	75	76	77	78
Length, km	1.00	0.75	0.75	0.50	0.50	1.05
Branch no.	79	80	81	82	83	84
Length, km	0.50	0.65	0.75	0.45	0.50	0.40
Branch no.	85	86	87	88	89	90
Length, km	2.10	0.75	0.75	0.50	0.50	1.05
Branch no.	91	92	93	94	95	96
Length, km	0.75	0.75	0.5	2.10	1.65	2.75
Branch no.	97					
Length, km	1.75					

TABLE A.III

Branch no.	Conductor mm ² /mm ²	Loading capacity A	Impedance Ω /km	Failure rate fl./km yr	Repair Duration h
1,2,3, 22,30,43,4 4,45,64,72	25/4	90	$1.2+j0.4$	0.2	3
Remaining	16/2.5	125	$2.1+j0.4$	0.2	3

TABLE A.IV
COST AND COMPLEMENTARY LOAD DATA

Power factor	Load factor	c_i US\$/kWh	c_l US\$/kWh	c_k kUS\$/km	g
0.9	0.6	4	0.1	15	0.05

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